2015-2016 – Econ590
Topics in Macroeconomics

Using VARs to Identify News Shocks

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Disclaimer

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1. How to observe news shocks ?

- Expectations (and expectation revisions) are not in the informational set on the econometrician
- Two possibilities :
  1. Narrative approach : looking at episodes
  2. More systematic approach : exploit some variability in the data.

2. Narrative Approach

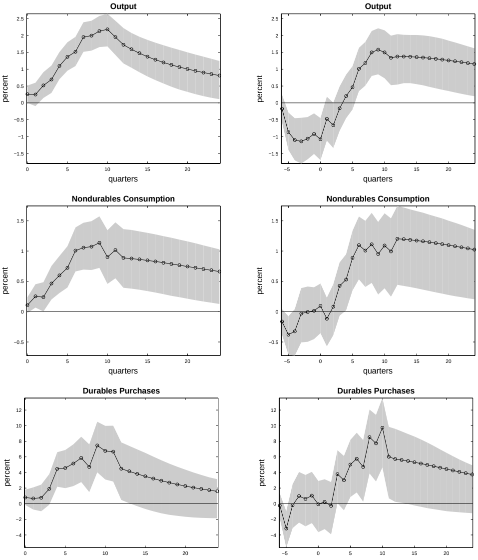
- Take an episode that is known as changing agents expectations, and then look at the consequences (First Gulf war (Blanchard aer-1998, Maastricht treaty ratification (Portier wp-2005)
- Nice recent work of Mertens & Ravn wp-2008 on announced *versus* surprising U.S. tax shocks

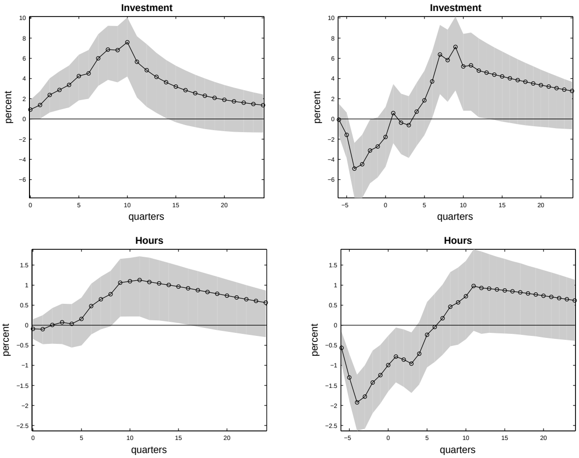
2. Narrative Approach

Mertens & Ravn aer-2012, ej-2011, red-2011

- Romer & Romer aer-2010 identify 49 significant legislated federal tax acts in the period 1947-2006 and a total of 104 separate changes in tax liabilities.
- For each of this change, one can observe the implementation lag (*"difference in timing between dates at which the tax legislations were signed by the President and the dates at which the tax liability changes were to be implemented"*
- 50% of episodes are anticipated and the median implementation lag of these legislations is 6 quarters.
- Results : - Agents do react to anticipated tax cut - They do so in in different manner compared to unexpected ones

Unanticipated Tax Shock   Anticipated Tax Shock




Investment
percent
quarters
Investment
percent
quarters
Hours
percent
quarters
Hours
percent
quarters

3. Systematic Approach

- A more *systematic* approach is a structural VAR one
- (Beaudry & Portier aer-2007) The idea is to look at some variables that capture agents expectations
- Typically asset prices.
- Let's restrict to possible information about future productivity.
- Do we find in asset prices data news about future total factor productivity (TFP) ? How does the economy react to those news ?

4. 2-VAR

- Simple bi-variate system.
- Measured total factor productivity TFP_t, and a forward looking economic decision variable X_t.
- The only characteristic of X_t that is important for our argument is that it be a jump variable, that is, a variable that can immediately react to changes in information without lag (stock price, interest rate, even consumption).

4. 2-VAR

- We look at the response of the TFP and the stock market X to a long run shock to TFP ("the" technology shock, called $\widetilde{\epsilon}_1$)
- We look at the response of the TFP and the stock market X to a stock market innovation ("the" non-technology shock, called ϵ_2)
- Standard theory suggest $\epsilon_2 \perp \widetilde{\epsilon}_1$.
- We show that data suggest $\epsilon_2 \; = \; \widetilde{\epsilon}_1$.

- We consider two alternative representations of the Bivariate VAR with orthogonalized errors :

$$\begin{pmatrix} \Delta TFP_t \\ \Delta X_t \end{pmatrix} = \Gamma(L) \begin{pmatrix} \epsilon_{1,t} \\ \epsilon_{2,t} \end{pmatrix}, \qquad (1)$$

$$\begin{pmatrix} \Delta TFP_t \\ \Delta X_t \end{pmatrix} = \tilde{\Gamma}(L) \begin{pmatrix} \tilde{\epsilon}_{1,t} \\ \tilde{\epsilon}_{2,t} \end{pmatrix}, \qquad (2)$$

- short run restriction : $\Gamma_{0(1,2)} = 0$ (ϵ_2 has no contemporaneous impact on TFP)
- long run restriction : $\tilde{\Gamma}(1)_{(1,2)} \left[= \sum_{i=0}^{\infty} \tilde{\Gamma}_i \right] = 0$ ($\tilde{\epsilon}_2$ has no long run impact on TFP)

4. 2-VAR

Prediction of a simple RBC model with technology and preference shocks

$$U = E_0 \sum_{t=0)}^{\infty} \beta^t \left[\log C_t - \Lambda_t \frac{L_t^\sigma}{\sigma} \right]$$

$$\Lambda_t = e^{\eta_{2,t}}$$

$$K_{t+1} = I_t$$

$$Y_t = \theta_t K_t^\gamma L_t^{1-\gamma}$$

$$\theta_t = \theta_{t-1} \; e^{\eta_{1,t}}$$

- The model solution can be written as

$$\begin{pmatrix} \Delta TFP_t \\ \Delta p_t^b \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ (1-\gamma)\frac{1}{1-\gamma L} - 1 & -\frac{(1-L)(1-\gamma)^2}{\sigma(1-\gamma L)} \end{pmatrix} \begin{pmatrix} \eta_{1,t} \\ \eta_{2,t} \end{pmatrix} \tag{1}$$

- This model implies $\epsilon_1 = \eta_1$, $\epsilon_2 = \eta_2$, $\widetilde{\epsilon}_1 = \eta_1$ and $\widetilde{\epsilon}_2 = \eta_2$
- In particular, this type of model implies that $\epsilon_2 \perp \widetilde{\epsilon}_1$.

- Small deviation from the RBC model
- *TFP* has both a permanent component – $\bar{\theta}_t$ – and a temporary component – ν_t, and we disregard preference shock.
- The important additional assumption is that permanent innovation to technology are known to agents 1 period before they actually impact TFP.

$$
\begin{aligned}
TFP_t &= \bar{\theta}_t + \nu_t \\
\bar{\theta}_{t+1} &= \bar{\theta}_t + \eta_{1,t} \\
\nu_t &= \rho \nu_{t-1} + \eta_{2,t}, \qquad 0 < \rho < 1
\end{aligned}
$$

- The model solution can be written as

$$\begin{pmatrix} \Delta TFP_t \\ \Delta p_t^b \end{pmatrix} = \begin{pmatrix} L & \frac{\gamma(1-L)}{(1-\rho L)} \\ \frac{(1-\gamma)L}{1-\gamma L} - 1 & \frac{(-\rho)(1-L)}{(1-\rho L)} \end{pmatrix} \begin{pmatrix} \eta_{1,t} \\ \eta_{2,t} \end{pmatrix} \quad (2)$$

- This model implies $\epsilon_1 = \eta_2$, $\epsilon_2 = \eta_1$, $\widetilde{\epsilon}_1 = \eta_1$ and $\widetilde{\epsilon}_2 = \eta_2$
- In particular, we have that ϵ_2 is co-linear to $\widetilde{\epsilon}_1$.

- Important aspect of the two first models : business cycle fluctuations can be decomposed into structurally meaningful supply driven and demand driven components.
- Demand is orthogonal to TFP and has no long run effect.

- The news model is different.
- Before supply shocks actually materialize, the economy already incorporates it.
- Such a model does not validate a decomposition in terms of demand versus supply effect.
- In the short run, an anticipated technological improvement (a news shock) looks like a demand effect, while in the long run it looks like a supply effect.

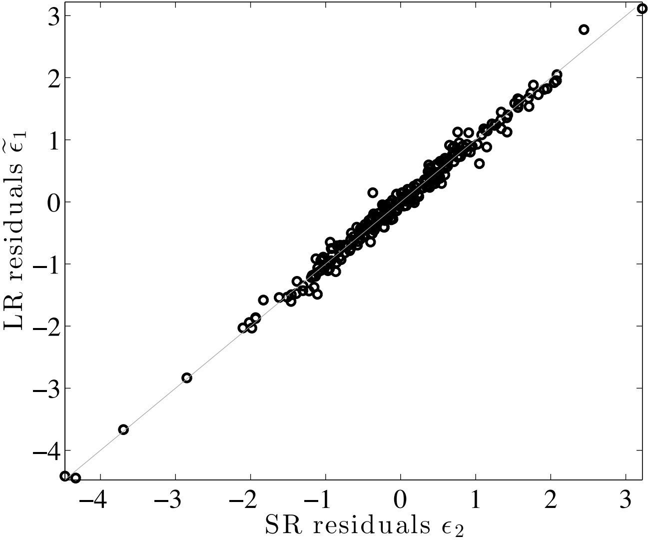
- 1947-2012 US data

4. 2-VAR

- We model the joint behavior of TFP and the stock price index.
- We obtain IRFs and epsilons

4. 2-VAR

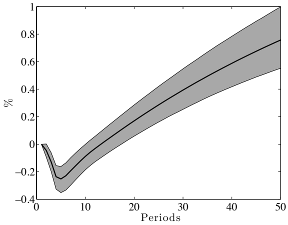
ϵ_2 Against $\widetilde{\epsilon}_1$ – U.S.



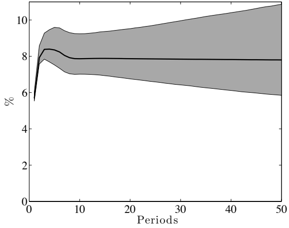
4. 2-VAR

Impulse Responses to Shocks ϵ_2 and $\widetilde{\epsilon}_1$ – U.S

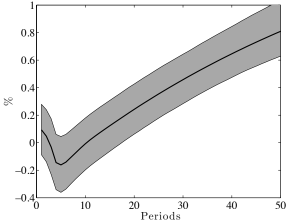
(a) *TFP*, response to ϵ_2



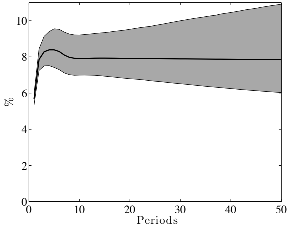
(b) *SP*, response to ϵ_2



(c) *TFP*, response to $\widetilde{\epsilon}_1$



(d) *SP*, response to $\widetilde{\epsilon}_1$



- Wold :

$$\begin{pmatrix} \Delta TFP_{1t} \\ \Delta SP_{2t} \\ \Delta X_{3t} \\ \vdots \\ \Delta X_{nt} \end{pmatrix} = C(L) \begin{pmatrix} \mu_{1,t} \\ \mu_{2,t} \\ \mu_{3,t} \\ \vdots \\ \mu_{nt} \end{pmatrix}.$$

- Structural :

$$\begin{pmatrix} \Delta TFP_{1t} \\ \Delta SP_{2t} \\ \Delta X_{3t} \\ \vdots \\ \Delta X_{nt} \end{pmatrix} = \Gamma(L) \begin{pmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \\ \varepsilon_{3,t} \\ \vdots \\ \varepsilon_{nt} \end{pmatrix}.$$

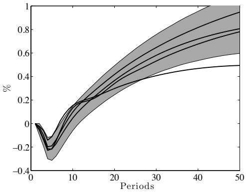
- To fully identify the n structural innovations, we need $n(n-1)/2$ restrictions.
- Consider a subset of restrictions that will allow us to identify only the news shock and a surprise technology shock.
  - × Only two shocks can permanently affect *TFP* in the long run
  - × $\rightsquigarrow$ $n-2$ zeros for the last $n-2$ columns of the first line of the long run matrix $\Gamma(1)$
  - × the surprise technology shock is the only shock that affects *TFP* on impact
  - × $\rightsquigarrow$ the only non-zero term in the first row of the impact matrix Γ_0 is the $(1,1)$ term.

These $2n-3$ restrictions allows for a unique identification of the news shock ε_2 and the surprise technology shock ε_1.

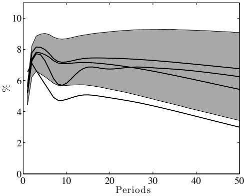
5. Systematic Approach : 3-VAR

A news shock triggers an expansion

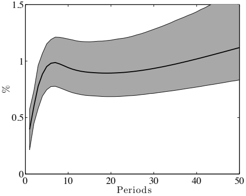
(a) TFP



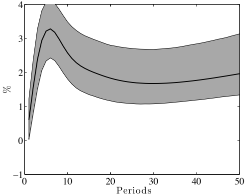
(b) Stock Prices



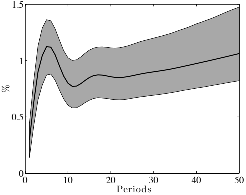
(c) Consumption



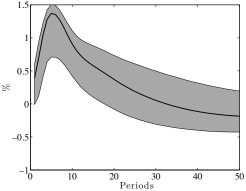
(d) Investment



(e) GDP

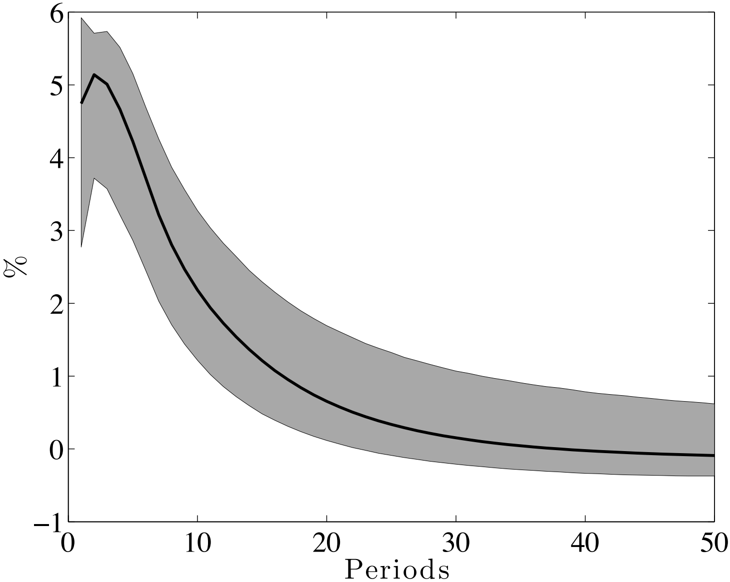


(f) Hours



5. Systematic Approach : 3-VAR

Consumer Confidence



5. Systematic Approach : 3-VAR

Share of the variance explained by the news shock in 3-variable VARs (in %)

Horizon	1	4	8	12	16	120
TFP	0	2	2	2	3	56
C	39	71	77	77	75	67
I	16	53	65	66	65	57
Y	30	72	82	80	78	64
H	39	70	81	82	79	56

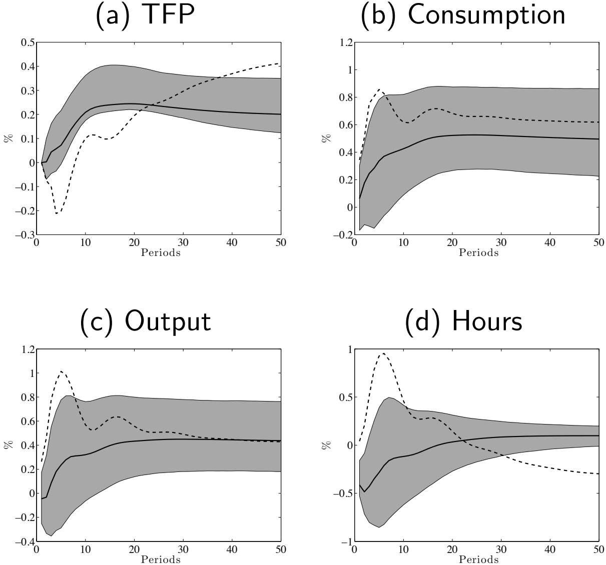
6. Other Approaches

Robustness

- Use different mix of short rune restrictions
- Use sign restrictions
- One particular approach : Barsky-Sims
  - × identify a surprise technology shock as the only one that affects TFP on impact.
  - × This shock would correspond to the first shock in a Choleski identification in which $X_1 = TFP$.
  - × Out of the $n-1$ other shocks, the news shock ε_2 is then identified as the one that maximizes $\sum_{j=1}^{h} \Omega_2(j)$, where $\Omega_2(j)$ is the forecast error variance of TFP at horizon j that is attributed to ε_2.
  - × $h = 40$
- We believe expansionary effect of tech. news shocks is robust.
- It is key to keep SP in the system.

6. Other Approaches

IRF to a news, Barsky-Sims identification in the (TFP, C, Y, H) VAR (dashed lines) and in the (TFP, SP, Y, H or C) one



- Suppose that the solution to a model generating the data is the one-dimensional stationary stochastic process x_t with moving average (MA) representation given by :

$$\begin{aligned} x_t &= \theta_0 \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_k \varepsilon_{t-k} + \cdots \\ &= \theta(L)\varepsilon_t. \end{aligned} \tag{3}$$

- **Definition :** A shock ε_t is said to be a *fundamental shock* if it can be recovered from observing the current and past observations of x.
- $\varepsilon_t = [\theta(L)]^{-1} x_t$
- $\rightsquigarrow$ All the roots of $\theta(L)$, *i.e.* the solution to the equation $\theta(z) = 0$, must be strictly outside the unit disc.

7. The Nonfundamentalness Problem

Problem

- When the shock in (3) is non-fundamental, the econometrician will typically not be able to recover the ε shocks from estimating the Wold representation
- The Wold representation is the unique linear representation of a stationary process where the shocks are linear forecast errors.
- One cannot recover the structural impulse response of x_t to a shock ε_t.

- **Definition :** a *news-rich* process as process for which there exists at least one q such that $|\theta_q| > |\theta_0|$.
- Such processes are of the "news" type in the sense that a larger share of the variance of x_t is attributable to the shock ε_{t-q} than to the shock ε_t, that is, more variance is due to a shock known q period in advance than due to the current period shock.
- Structural models with news shocks generally have a solution that correspond to a *news-rich* process.

- Model :

$$
\begin{aligned}
x_t &= aE_t x_{t+1} + y_t, \\
y_t &= \varepsilon_t + (1+\alpha)\varepsilon_{t-1},
\end{aligned}
$$

  with $0 < a < 1$, $E(\varepsilon_t) = 0$ and $E(\varepsilon_t^2) = \sigma^2$.
- x_t is an endogenous variable and y_t is the news-rich exogenous process.
- Solving this model forward, we obtain

$$
x_t = \sum_{\tau=0}^{\infty} a^{\tau} E_t y_{t+\tau}.
$$

- Using the process of y, the model's solution reduces to

$$
x_t = (1 + a(1+\alpha))\varepsilon_t + (1+\alpha)\varepsilon_{t-1}.
$$

7. The Nonfundamentalness Problem

An Example

- $$x_t = (1 + a(1+\alpha))\varepsilon_t + (1+\alpha)\varepsilon_{t-1}.$$
- The solution to the model is therefore a news-rich process if $a < \frac{\alpha}{1+\alpha}$.
- Moreover, this is the same condition that implies non-fundamentalness, as the root of $\theta(z)$ will be inside the unit circle if and only if $a < \frac{\alpha}{1+\alpha}$.
- This simple example shows
  - *(i)* that news shocks ($\alpha > 0$) do not necessarily mean non-fundamentalness
  - *(ii)* that if the model solution is a news-rich MA(1), , the ε_ts are not fundamental shocks and therefore cannot be recovered from current and past observations on x_t.
  - In other words, one cannot use the Wold representation of the data to obtain structurally meaningful impulse responses.

7. The Nonfundamentalness Problem

Is the fundamental representation close to the non-fundamental representation in the MA(1) case ?

- Assume the news-rich process

$$x_t = \varepsilon_t + (1+\alpha)\varepsilon_{t-1}$$

- Wold representation :

$$x_t = \widetilde{\varepsilon}_t + \theta_1 \widetilde{\varepsilon}_{t-1},$$

  where θ_1 and $\widetilde{\sigma}^2$ need to be found
- We can obtain those two parameters by matching the autocorrelation functions. Computing variance and autocorrelation :

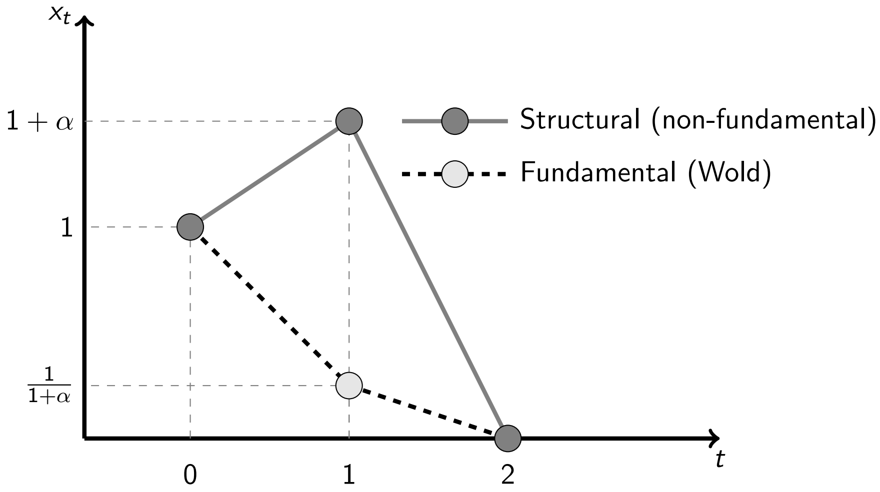
$$\begin{cases} E(x_t^2) & = & (1+(1+\alpha)^2)\sigma_\varepsilon^2 & = & (1+\theta_1^2)\sigma_{\widetilde{\varepsilon}}^2, \\ E(x_t x_{t-1}) & = & (1+\alpha)\sigma_\varepsilon^2 & = & \theta_1 \sigma_{\widetilde{\varepsilon}}^2. \end{cases}$$

- Solving those two equations, we obtain the following fundamental representation

$$x_t = \widetilde{\varepsilon}_t + \frac{1}{1+\alpha}\widetilde{\varepsilon}_{t-1}.$$

7. The Nonfundamentalness Problem

Is the fundamental representation close to the non-fundamental representation in the MA(1) case ?



- Consider the MA(1) news-rich process :

$$x_t = \theta_0 \varepsilon_t + (\theta_0 + \alpha)\varepsilon_{t-1},$$

  with (for simplicity) $\theta_0 \geq 0$ and $\alpha > 0$.
- ε is non-fundamental, more precisely x_t-non-fundamental
- Let's give some more information to the econometrician (an "asset price") :

$$y_t = \gamma \varepsilon_t + \nu_t$$

  whereν_t is noise.

- We have the bi-variate model

$$\begin{aligned} x_t &= \theta_0 \varepsilon_t + (\theta_0 + \alpha)\varepsilon_{t-1} + \nu_t, && (4) \\ y_t &= \gamma \varepsilon_t + \nu_t, && (5) \end{aligned}$$

  with $\gamma > 0$ for simplicity.
- Matrix representation is

$$\begin{pmatrix} x_t \\ y_t \end{pmatrix} = \begin{pmatrix} \theta_0 + (\theta_0 + \alpha)L & 1 \\ \gamma & 1 \end{pmatrix} \begin{pmatrix} \varepsilon_t \\ \nu_t \end{pmatrix} = \Theta(L) \begin{pmatrix} \varepsilon_t \\ \nu_t \end{pmatrix}.$$

$$\begin{pmatrix} x_t \\ y_t \end{pmatrix} = \begin{pmatrix} \theta_0 + (\theta_0 + \alpha)L & 1 \\ \gamma & 1 \end{pmatrix} \begin{pmatrix} \varepsilon_t \\ \nu_t \end{pmatrix} = \Theta(L) \begin{pmatrix} \varepsilon_t \\ \nu_t \end{pmatrix}.$$

- The process will be invertible, and therefore ε will be fundamental, when the roots of the determinant of the matrix $\Theta(z)$ are outside the unit disk.
- Determinant of $\Theta(z)$ has only one root which is $z = \frac{\gamma - \theta_0}{\theta_0 + \alpha}$.
  - × If $\gamma = 0$, meaning that y is uninformative about ε, then $|z| < 1$ and ε is non-fundamental
  - × But if y is informative enough about ε, meaning that γ is large enough (precisely $\gamma > 2\theta_0 + \alpha$), then ε is fundamental.

- Consider an economic model who has a representation for $\{y_t\}$ in the state-space form

$$\begin{aligned} x_{t+1} &= Ax_t + B\varepsilon_t, & (6) \\ y_t &= Cx_{t-1} + D\varepsilon_t. & (7) \end{aligned}$$

- x_t is $n \times 1$ vector of possibly unobserved state variables, y_t is a $k \times 1$ vector of variables observed by the econometrician, and ε_t is $m \times 1$ vector of structural shocks.
- The shocks ε are Gaussian vector white noise with $E\varepsilon_t = 0$, $E\varepsilon_t\varepsilon_t' = I$ and $E\varepsilon_t\varepsilon_{t-j} = 0$ for $j \neq 0$.
- We restrict to the "square case" in which there are as many shocks as observables in y ($k = m$) and in which D has full rank.

- The question we ask is whether or not one can recover the structural shocks from the following VAR that involves only the observable variables y :

$$y_t = F(L)y_{t-1} + G\widehat{\varepsilon}_t, \tag{8}$$

where $F(L)$ is a infinite order lag polynomial.

- In other words, does there exist an $F(L)$ and a matrix G such that $\widehat{\varepsilon}_t = \varepsilon_t$?

- Note that the answer to the question relies on two different sets of properties.
- First, assuming perfect observability of the state variables ($y = x$), the question is whether the VAR process (6) is invertible, which is a constraint on A.
- Second, given that information is imperfect, is the informational content of y enough to recover the structural shocks, which is a constraint on C and D given A and B.

- Solve for ε_t in (7) :

$$\varepsilon_t = D^{-1}(y_t - Cx_{t-1}),$$

- then plug into (6) :

$$x_t = (A - BD^{-1}C)x_{t-1} + BD^{-1}y_t.$$

- Solving backward yields

$$x_t = (A - BD^{-1}C)^{t-1}x_0 + \sum_{\tau=0}^{t-1}(A - BD^{-1}C)^{\tau-1}BD^{-1}y_{t-\tau}. \quad (9)$$

- If $\lim_{t\to\infty}(A - BD^{-1}C)^{t-1} = 0$ (which is the case if $(A - BD^{-1}C)$ is a stable matrix), then the history of observables perfectly reveals the current state. One can then plug (9) in (7) to obtain a VAR in obervables whose innovations are ε :

$$y_t = C\sum_{\tau=0}^{t-1}(A - BD^{-1}C)^{\tau-1}BD^{-1}y_{t-1-\tau} + D\varepsilon_t.$$

$$x_t = (A - BD^{-1}C)^{t-1}x_0 + \sum_{\tau=0}^{t-1}(A - BD^{-1}C)^{\tau-1}BD^{-1}y_{t-\tau}.$$

- **Poor Man's Invertibility Condition** : $(A - BD^{-1}C)$ is a stable matrix.
- Then $\lim_{t\to\infty}(A - BD^{-1}C)^{t-1} = 0$ and the history of observables perfectly reveals the current state. One can then plug (9) in (7) to obtain a VAR in observables whose innovations are ε :

$$y_t = C\sum_{\tau=0}^{t-1}(A - BD^{-1}C)^{\tau-1}BD^{-1}y_{t-1-\tau} + D\varepsilon_t.$$

- The structural shocks ε are therefore fundamental.

- If $A - BD^{-1}C$ is not stable, then the structural shocks cannot be obtained from the estimation of a VAR like (8).
- In such a case, how close will be the fundamental shocks to the structural one depends on how much information is contained in y.
- use the Kalman filter to form a forecast of the current state, $\widehat{x}$, given observables and a lagged forecast :

$$\widehat{x}_t = (A - KC)\widehat{x}_{t-1} + Ky_t, \tag{10}$$

where K is the time invariant Kalman gain and $A - KC$ a stable matrix under some stabilizability and detectability conditions

- Now

$$y_t = C\widehat{x}_{t-1} + u_t, \tag{11}$$

with

$$u_t = C(x_{t-1} - \widehat{x}_{t-1}) + D\varepsilon_t. \tag{12}$$

- Using lagged (10), substituting in (11) and solving backward gives, when $t \to \infty$ and using the fact that $(A - KC)$ is a stable matrix :

$$y_t = C\sum_{\tau=0}^{t-1}(A - KC)^{\tau} K y_{t-1-\tau} + u_t. \tag{13}$$

$$y_t = C \sum_{\tau=0}^{t-1} (A - KC)^{\tau} K y_{t-1-\tau} + u_t.$$

- This is a VAR, but with "wrong" (nonstructural) residuals

$$\Sigma_u = C \widetilde{\Sigma} C' + DD',$$

$$\Sigma_u = C\widetilde{\Sigma}C' + DD',$$

- When the poor man's invertibility condition holds, $\widetilde{\Sigma} = 0$, so that the structural shock is fundamental.
- When that condition does not hold, $\widetilde{\Sigma} \neq 0$, and the innovation variance from the VAR is strictly larger than the innovation variance in the structural model.
- Invertibility fails because the observables do not allow for full revelation of the state vector.
- Put that way, it becomes clear that non-invertibility is fundamentally an issue of missing information